SCIENCE OLYMPIAD FOUNDATION

Class 8

IMO

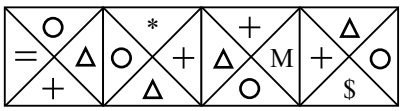
Set B

Year 2017

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1. Select a figure from the options which will continue the same series as established by the Problem Figures.

Problem Figures



A. 

B. 

C. 

D. 

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2. In a certain code language, the word VICTORY is written as TKAVMTW. How will the word SUCCESS be written in that code language?

A. USEAGQU
B. QSACGUU
C. UWECCQQ
D. QWAECUQ

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3. Select a figure from the options which shows the correct mirror image of Fig. (X), if the mirror is placed vertically to the right.

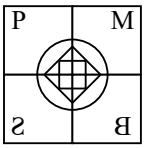
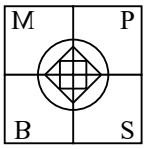
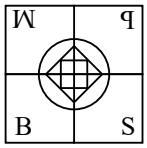
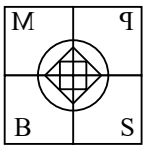
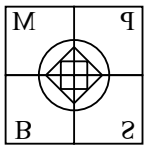


Fig. (X)

A. 

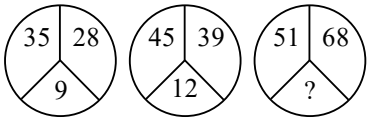
B. 

C. 

D. 

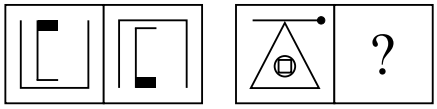
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4. Find the missing number, if the same rule is followed in all the three figures.



A. 35
B. 19
C. 17
D. 21

5. There is a certain relationship between figures (i) and (ii). Establish a similar relationship between figures (iii) and (iv) by selecting a suitable figure from the options which will replace the (?) in figure (iv).



(i) (ii) (iii) (iv)

A. 

B. 

C. 

D. 

---

6. Pointing to a woman in a photograph, a man said, "She is the only daughter of the only daughter of my brother-in-law's father". How is the woman related to the man?

A. Wife
B. Mother
C. Daughter
D. Niece

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7. Select a figure from the options which satisfies the same condition of placement of dots as in Fig. (X).

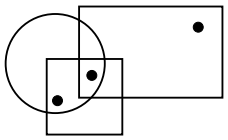
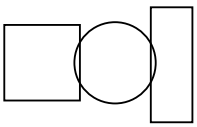
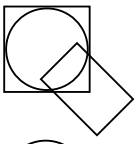
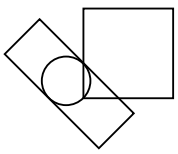
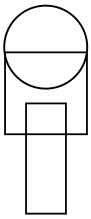


Fig. (X)

A. 

B. 

C. 

D. 

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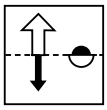
8. Some children A, B, C, D, E, F and G are standing in a line facing North. G is to the right of D and to the left of B. A is on the right of C. A and D have one child between them. E and B have two children between them. D and F have two children between them.

Who is exactly in the middle?

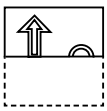
A. A
B. C
C. D
D. E

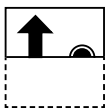
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9. A square transparent Sheet (X) with a pattern is given. Select a figure from the options as to how the pattern would appear when the transparent sheet is folded along the dotted line.

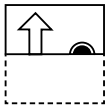


Sheet (X)

A. 

B. 

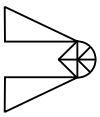
C. 

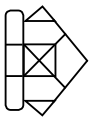
D. 

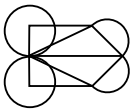
10. Select a figure from the options in which Fig. (X) is exactly embedded as one of its parts.

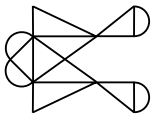


Fig. (X)

A. 

B. 

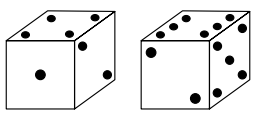
C. 

D. 

11. If '–' denotes '÷', '+' denotes '×', '÷' denotes '–' and '×' denotes '+', then what is the value of $43 \times 7 \div 5 + 8 - 4$?

A. 47
B. 40
C. 90
D. 45

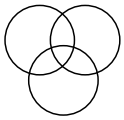
12. Two positions of a dice are shown here. When 4 dots are at the bottom, the number of dots at the top will be _____.



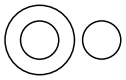
A. 3
B. 5
C. 6
D. 1

13. Which of the following Venn diagrams best illustrates the relationship amongst, "Rings, Ornaments, Diamond rings"?

A. 

B. 

C. 

D. 

14. The two rows of numbers are given. Resultant number in each row is to be worked out separately based on the following rules and the question below the row of numbers is to be answered. The operations of numbers progress from left to right.

Rules:

(i) If an even number is followed by a prime number, then they are to be multiplied.

(ii) If an even number is followed by a composite odd number, then the odd number is to be subtracted from the even number.

(iii) If a composite odd number is followed by a prime number, then the first number is to be divided by the second number.

(iv) If an odd number is followed by an even number which is a perfect square, then they are to be added.

(v) If an odd number is followed by another odd number, then they are to be added.

86	45	13	21
m	11	36	21

If *m* is the resultant of the first row, then what is the resultant of the second row?

A. 50
B. 77
C. 60
D. 43

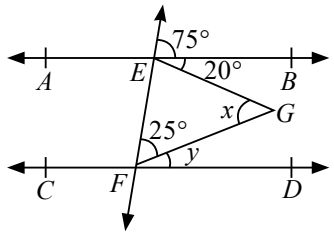
15. Which of the following numbers will be obtained if the second digit of the smallest number is added to the third digit of the greatest number after adding 5 to each number of the given arrangement?

152 453 279 536 785

A. 6
B. 7
C. 5
D. 4

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16. In the given figure, $AB||CD$ and $\overleftrightarrow{EF}$ is the transversal. Find the values of x and y respectively.



A. 70°, 30°
B. 70°, 50°
C. 50°, 30°
D. 50°, 70°

17. Numbers 1 to 10 are written on ten separate slips, kept in a box and mixed well. One slip is chosen from the box without looking into it. What is the probability of

(i) Getting a 1-digit number?
(ii) Getting a prime number?

	(i)	**(ii)**
A.	9/10	5/10
B.	9/10	4/10
C.	1/10	4/10
D.	5/10	3/10

18. If $\frac{9^n \times 3^2 \times \left[3^{\left(\frac{-n}{2}\right)}\right]^{-2} - (27)^n}{3^{3m} \times 2^3} = \frac{1}{27}$, then find the relationship between m and n.

A. $m + n = -1$
B. $m - n = 1$
C. $m = 1 - n$
D. $\frac{m}{n} = 1$

19. For two rational numbers $\frac{A}{B}$ and $\frac{C}{D}$, that are in standard form, which of the following conditions are true?

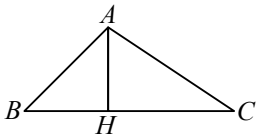
A. If $AD > BC$, then $\frac{A}{B} > \frac{C}{D}$.
B. If $AB < CD$, then $\frac{A}{B} < \frac{C}{D}$.
C. If $AB = CD$, then $\frac{A}{B} = \frac{C}{D}$.
D. None of these

20. Solve for x:

$$\frac{2}{5}(4x-1) - \left[4x - \left(\frac{1-3x}{2}\right)\right] = \frac{x-7}{2}$$

A. 0
B. 9/11
C. 2/11
D. –1

21. ABC is a right angled triangle with $\angle BAC = 90°$. AH is drawn perpendicular to BC. If $AB = 60$ cm and $AC = 80$ cm, then find the length of BH.

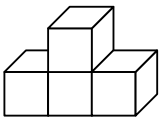


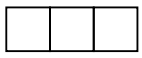
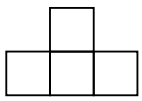
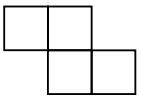
A. 36 cm
B. 32 cm
C. 24 cm
D. 30 cm

22. Which of the following statements is CORRECT ?

A. A unique quadrilateral can be constructed if its two sides and a diagonal is given.
B. A unique quadrilateral can be constructed if its two adjacent sides and two angles are given.
C. A unique quadrilateral can be constructed if its three sides and two included angles are given.
D. None of these

23. Which of the following is the top view of the given figure?



A. 
B. 
C. 
D. 

24. In two triangles ABC and DEF, if $AB = BC$, $\angle B = \angle E$, $DE = EF$ and $BC = EF$, then which of the following statements is CORRECT?

A. Triangles are congruent only.
B. Triangles are isosceles only.
C. Both the triangles are congruent and isosceles.
D. Both the triangles are neither congruent nor isosceles.

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25. If $\sqrt{0.04\times0.4\times a} = 0.004\times0.4\times\sqrt{b}$, then $\frac{a}{b}$ is equal to

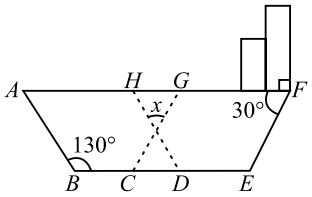
A. 16×10^{-3}
B. 16×10^{-4}
C. 16×10^{-5}
D. 16×10^{-2}

26. Simplify :

$$\frac{x^2-(y-z)^2}{(x+z)^2-y^2}+\frac{y^2-(x-z)^2}{(x+y)^2-z^2}+\frac{z^2-(x-y)^2}{(y+z)^2-x^2}$$

A. –1
B. 0
C. 1
D. 2

27. In the following figure *ABDH* and *CEFG* are two parallelograms. Find the value of *x*.



A. 120°
B. 100°
C. 50°
D. 30°

28. The difference between the compound interest (compounded annually) and simple interest on a certain sum at 10% per annum for 2 years is ₹ 631. Find the sum.

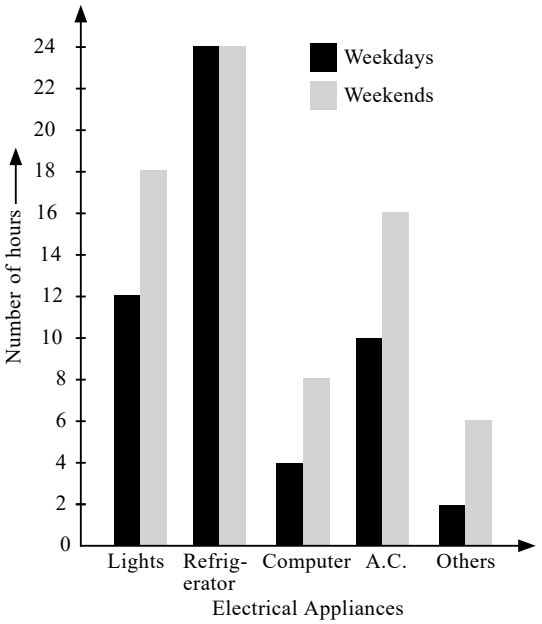
A. ₹ 80100
B. ₹ 63100
C. ₹ 60000
D. ₹ 73100

29. Given below are the steps of construction of a quadrilateral *ABCD*, where *AB* = 3.5 cm, *BC* = 6.5 cm, $\angle A = 75°$, $\angle B = 105°$ and $\angle C = 120°$.Which of the following steps is INCORRECT?

Step 1 : Draw *AB* = 3.5 cm.
Step 2 : Draw $\angle XAB = 75°$ at *A* and $\angle ABY = 105°$ at *B*.
Step 3 : With *B* as centre and radius *BC* = 6.5 cm, draw an arc to intersect *BY* at *C*.
Step 4 : At *C*, draw $\angle ADC = 120°$ such that *CZ* meets *AX* at *D*.

A. Step 1 only
B. Step 2 only
C. Step 3 only
D. Step 4 only

Direction (30-31) : The given double column graph compares the average number of hours for which an electrical appliance is used on weekdays and weekends. Study the graph carefully and answer the questions that follows.



30. Which appliance is used half as many hours on the weekdays as it is used on the weekends?

A. Lights
B. Refrigerator
C. Computer
D. A.C.

31. What is the total number of appliances used in weekdays?

A. 82
B. 72
C. 154
D. None of these

32. The given table shows the cost price of various items purchased from a showroom.

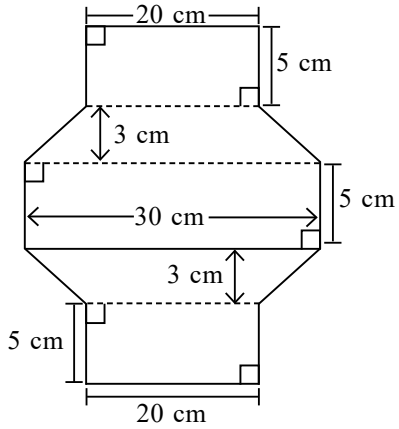
Items	Cost (in ₹)
Shirt	1250
Trouser	1550
2 T-shirts	2200

What will these items cost, if the GST is 12%?

A. ₹ 5350
B. ₹ 5600
C. ₹ 5000
D. ₹ 5750

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33. Find the area of the given figure (not drawn to scale).



A. 650 cm^2
B. 500 cm^2
C. 575 cm^2
D. 525 cm^2

34. If $\left(x^2+\frac{1}{x^2}\right)=34$, $(x > 0)$, then find the value of $x^3+\frac{1}{x^3}$.

A. 198 B. 216
C. 200 D. 186

35. If $n = 1 + x$ and x is the product of four consecutive integers, then which of the following is true?

I. n is an odd integer.
II. n is prime.
III. n is a perfect square.

A. Only I is correct.
B. Only III is correct.
C. Both I and II are correct.
D. Both I and III are correct.

EVERYDAY MATHEMATICS

36. The ratio of a man's age to his son's age is 4 : 1. The product of their ages is 196. What will be the ratio of their ages after 5 years?

A. 3 : 1
B. 10 : 3
C. 11 : 4
D. 14 : 5

37. Mohit went to a shop to buy 50 kg of rice. He buys two varieties of rice which cost him ₹ 4.50 per kg and ₹ 5.00 per kg. He spends a total of ₹ 240. What was the quantity of less expensive rice bought by him?

A. 20 kg
B. 25 kg
C. 30 kg
D. 35 kg

38. Ashima has three cylindrical containers.

(i) P having radius $2r$ and height h.
(ii) Q having radius $3r$ and height $2h$.
(iii) R having radius $3r$ and height $\frac{h}{3}$.

Arrange the curved surface area of containers in decreasing order.

A. Q, P, R
B. P, R, Q
C. P, Q, R
D. Q, R, P

39. One custard recipe requires $\frac{1}{2}$ cup of sugar. Another recipe for the same custard requires 2 table-spoons of sugar. If 1 table-spoon is equivalent to $\frac{1}{16}$ cup, then how much more sugar does the first recipe require?

A. $\frac{4}{5}$ cup B. $\frac{6}{5}$ cup
C. $\frac{3}{8}$ cup D. $\frac{5}{8}$ cup

40. The cost of papering the four walls of a room is ₹ 480. Each one of the length, breadth and height of another room is double that of this room. What is the cost of papering the walls of the new room?

A. ₹ 3840
B. ₹ 2880
C. ₹ 1920
D. ₹ 960

41. A train 270 m long is moving with a speed of 25 km/hr. In what time will it cross a man coming from the opposite direction with a speed of 2 km/hr in which the train is moving?

A. 36 sec
B. 32 sec
C. 28 sec
D. 24 sec

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42. Harshad bought 15 pieces of DVD players at ₹ 4500 each and sold all of them at the total price of ₹ 81000. What is the percent profit earned in the deal?

A. $16\frac{2}{3}$ % B. 20 %

C. $20\frac{1}{2}$ % D. 25 %

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43. An urn contains 2 red, 3 green and 2 blue balls. If a ball is drawn at random, then find the probability that the ball is not blue.

A. $\frac{5}{7}$ B. $\frac{10}{21}$

C. $\frac{2}{7}$ D. $\frac{11}{21}$

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44. A General of army wants to arrange 36562 soldiers in equal number of rows and columns. After the arrangement, he found that some soldiers were left. How many soldiers were left?

A. 36
B. 65
C. 81
D. 97

---

45. A shopkeeper offers 3.5% discount on cash purchases. What cash amount would Trishi pay for a dress, if the marked price of the dress is ₹ 1250?

A. ₹ 1204.75
B. ₹ 1205
C. ₹ 1206
D. ₹ 1206.25

ACHIEVERS SECTION

46. Which of the following options is CORRECT?

A. Additive identity of a rational number a is $-1/a$.

B. For rational numbers a, b and c, $a - (b - c) = (a - b) - c$.

C. The value of $\dfrac{1}{1+\dfrac{2/3}{1+\dfrac{2}{3}+\dfrac{8/9}{1-2/3}}}$ is $\frac{13}{15}$.

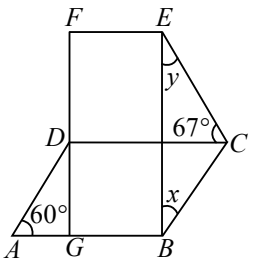
D. Reciprocal of $\frac{1}{x}$, (where $x \neq 0$) is $-x$.

---

47. In the given figure (not drawn to scale), $ABCD$ is a parallelogram and $GBEF$ is a rectangle. Find

(i) x

(ii) y



	(i)	(ii)
A.	60°	47°
B.	30°	23°
C.	30°	67°
D.	60°	30°

48. Read the statements carefully and select the CORRECT option.

Statement-1 : While solving a mathematical problem, Samidha squared a number and then subtracted 25 from it rather than the required *i.e.*, first subtracting 25 from the number and then squaring it. But she got the right answer. The given number is 13.

Statement-2 : The smallest natural number which is a perfect square and which ends in 3 identical digits lies between 2000 and 3000.

A. Statement-1 is true but Statement-2 is false.
B. Statement-1 is false but Statement-2 is true.
C. Both Statement-1 and Statement-2 are true.
D. Both Statement-1 and Statement-2 are false.

---

49. Match the following.

	Column-I		Column-II
(P)	$(x^2 - 3x + 2) \times (x^2 + 2x + 1)$	(i)	$3x^3 - 4x^2 - 12x + 8$
(Q)	$(2x^2 - 3x + 1) + (3xy - 4x^2 - 2x + 5)$	(ii)	$x^4 - x^3 - 3x^2 + x + 2$
(R)	$(2x^3 - 3x^2 - 7x + 6) - (x^2 - x^3 + 5x - 2)$	(iii)	$2x - 2y$
(S)	$(4x^2 + 4y^2 - 8xy) \div (2x - 2y)$	(iv)	$-2x^2 + 3xy - 5x + 6$

	(P)	(Q)	(R)	(S)
A.	(ii)	(i)	(iii)	(iv)
B.	(ii)	(iv)	(i)	(iii)
C.	(iii)	(i)	(ii)	(iv)
D.	(ii)	(iv)	(iii)	(i)

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50. State T for true and F for false and select the CORRECT option.

(i) Compound interest is calculated by using the formula, $\text{C.I.} = P\left(1+\frac{r}{100}\right)^n$

(ii) The simple interest and the compound interest on the same principal for same time period (> 1 year) and at the same rate of interest are not equal.

(iii) If a sum of ₹ 2000 becomes ₹ 2600 after three years, then rate of simple interest is 10% per annum.

(iv) In case of compound interest, the principal remains constant throughout the time period.

	(i)	**(ii)**	**(iii)**	**(iv)**
A.	T	T	F	F
B.	F	T	T	F
C.	F	T	F	T
D.	T	F	F	T

SPACE FOR ROUGH WORK

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